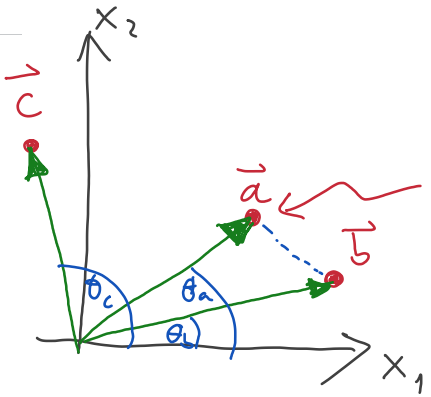
K-NN

Friday, September 18, 2026 8:20 AM

Vector $\vec{c}$



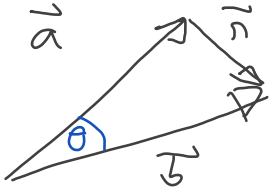
$\vec{a}$ ← Sample, observation

$\begin{bmatrix}1\\2\end{bmatrix} \gtrless \begin{bmatrix}2\\1\end{bmatrix}$ $1 < 2$

$\vec{a}+\vec{n}-\vec{b}=\vec{0}$

$\vec{n}=\vec{b}-\vec{a}$

$\|\vec{n}\|_2 = \|\vec{b}-\vec{a}\|_2$



$\frac{3!}{2!\,1!} = 3$

$*|\theta_a-\theta_b| = |\theta_b-\theta_a|$

$|\theta_b-\theta_c|$

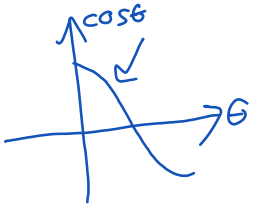
$|\theta_a-\theta_c|$

$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$ Euclidian

Cosine

$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

$\theta = \arccos\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$



$\vec{x} = \begin{bmatrix}1\\2\\3\end{bmatrix}$ ← Column Vector

$\vec{x} = [1 \quad 2 \quad 3]$ ← row vector

$\vec{x}_1 = [1, 0, 2]$ $y_1 = Y$

$\vec{x}_2 = [2, -1, -3]$ $y_2 = N$

$\vec{x}_3 = [3, 1, 0]$ $y_3 = N$

Tabular data

#	A_1	A_2	A_3	Target
1	1	0	2	Y
→ 2	2	-1	-3	N
→ 3	3	1	0	N
4	0	1	1	?

$\frac{\Delta A_1^2 + \Delta A_2^2 + \Delta A_3^2}{\text{Squared Euclidean}}$

$D = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 27 & 9 & 3 \\ 2 & 27 & 0 & 14 & 24 \\ 3 & 9 & 14 & 0 & 10 \\ 4 & 3 & 24 & 10 & 0 \end{array}$

$\|\vec{x}_1-\vec{x}_2\|^2 = 1+1+25 = 27$

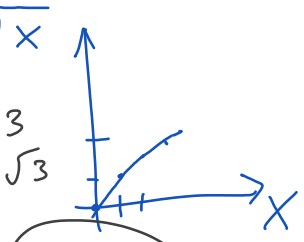
$\|\vec{x}_1-\vec{x}_3\|^2 = 4+1+4 = 9$

$\|\vec{x}_2-\vec{x}_3\|^2 = 1+4+9 = 14$

$\sqrt{x}$

$4 > 3$

$\sqrt{4} > \sqrt{3}$



$x \propto \sqrt{x}$

$D_{cose} = \begin{bmatrix} 1 & -0.48 & 0.42 & 0.63 \\ -0.48 & 1 & 0.42 & -0.76 \\ 0.42 & 0.42 & 1 & 0.22 \\ 0.63 & -0.76 & 0.22 & 1 \end{bmatrix}$

$\frac{\vec{1}\cdot\vec{2}}{\|\vec{1}\|\|\vec{2}\|} = \frac{2+0-6}{\sqrt{5}\sqrt{14}} = \frac{-4}{\sqrt{70}}$

$D = D'$ ① ② ③ ④

$\|\vec{x}_4 - \vec{x}\|^2 = 1+1+1 = 3$

$\|\vec{x}_4 - \vec{x}_2\|^2 = 4+4+16 = 24$

$\|\vec{x}_4 - \vec{x}_3\|^2 = 9+0+1 = 10$

k = 1 distance = 3 → ① → Y

k = 2 (3, 10) → (①, ③) → (Y, N)

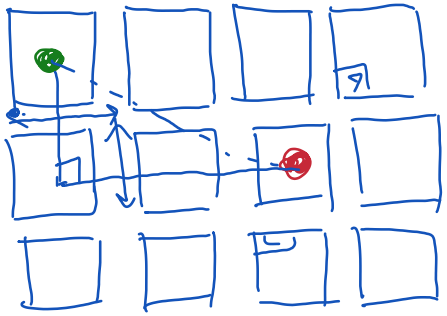
k = 3 (3, 10, 24) → (①, ③, ②) → (Y, N, N) → N

k = 1 0.63 → ① → Y

k = 2 (0.63, 0.22) → (①, ⑤) → (Y, N)

k = 3 (0.63, 0.22, -0.75) → (①, ③, ②) → (Y, N, N) → N

Manhattan Distance



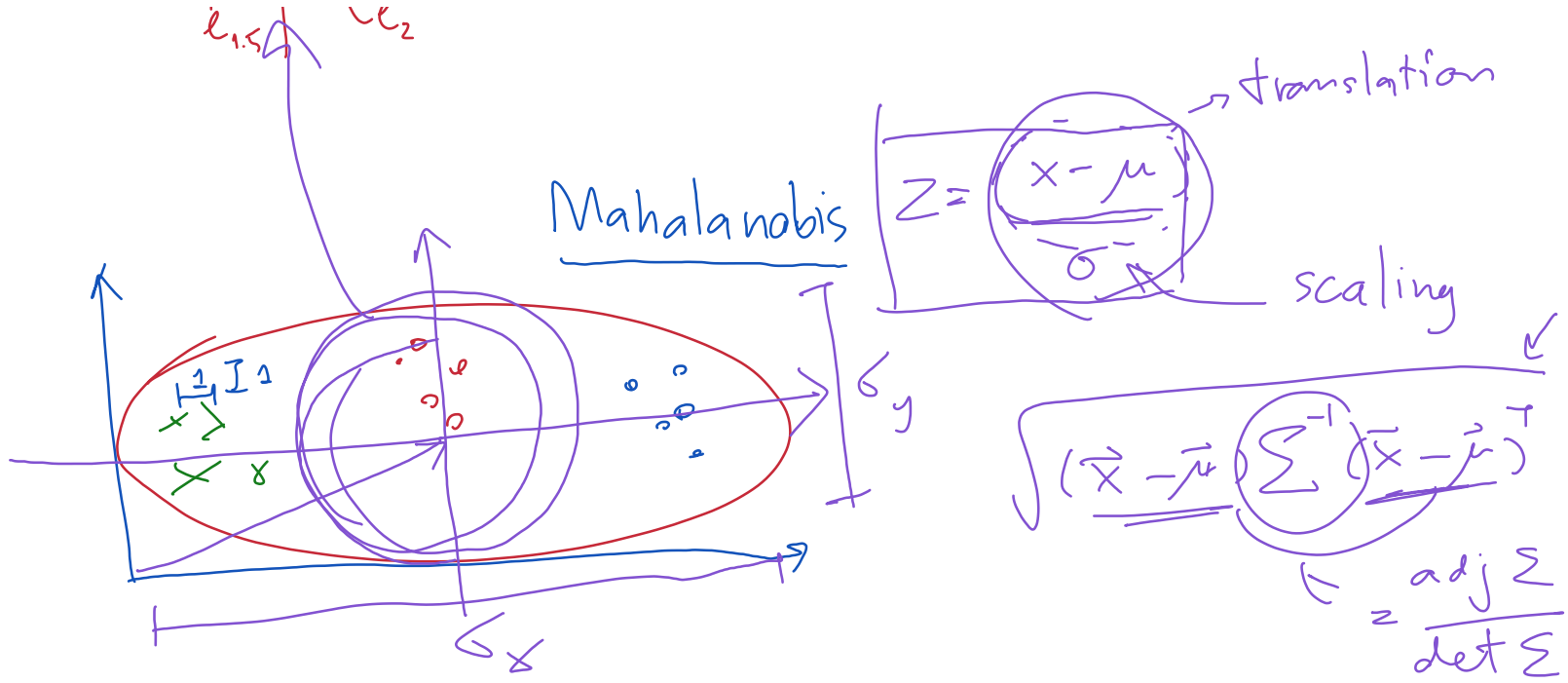
$\sqrt{|x_1-x_2|^2+|y_1-y_2|^2} = \ell_2\text{-norm} = \|\vec{a}\|_2$ → MSE

$|x_1-x_2|+|y_1-y_2| = \ell_1\text{-norm} = \|\vec{a}\|_1$ → MAE

ℓ_p-norm

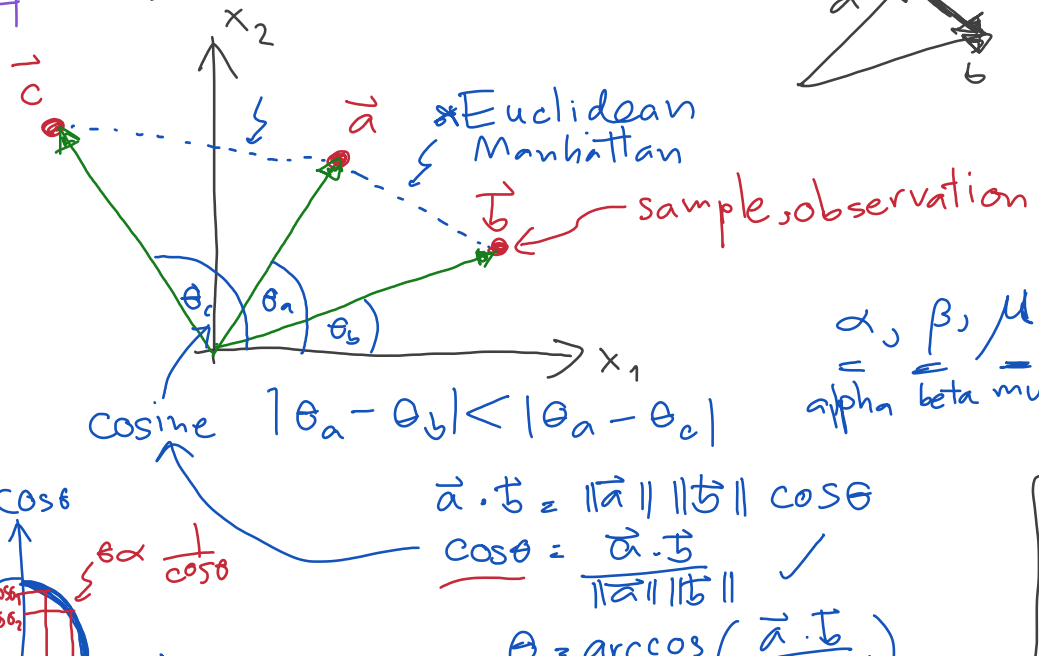
$\sqrt[p]{a_1^p + a_2^p + a_3^p} = \left(\sum_{i=1}^{n} a_i^p\right)^{\frac{1}{p}} = \|\vec{a}\|_p$ → 1.5, 0.5, 3

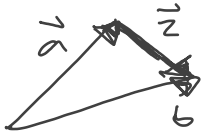




PM

k-NN





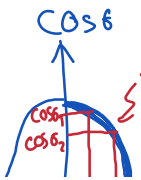
$\vec{a} + \vec{z} - \vec{b} = 0$

$\|\vec{z}\|_2 = \|\vec{b} - \vec{a}\|_2$

$\Sigma \quad \Pi \quad \Lambda \quad \Theta$

$\alpha, \beta, \mu, \eta, \sigma, \pi, \lambda, \theta$

alpha beta mu eta sigma Pi lambda Theta

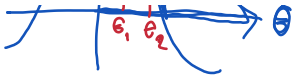


$\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos\theta$

$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|}$ ✓

$\theta = \arccos\left(\frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|}\right)$

$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} - \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$



$(\|a\|\|b\|)$

$$d = \sqrt{(a_1-b_1)^2 + (a_2-b_2)^2 + (a_3-b_3)^2}$$

$d \propto \sqrt{d}$ Squared Euclidean

Tabular data

#	A_1	A_2	A_3	Target
1	1	2	3	Y
2	3	2	1	N
3	2	1	1	N
4	1	1	2	?

training set / test set

k-NN
1-3
Squared Euclidean

k=1 (2, 2, 5) → (①, ③, ②) → (Y, N, N) → Y
k=2 → Y
k=3 → N

$$D_e = \begin{bmatrix} 1 & 0 & 8 & 6 & 2 \\ 2 & 8 & 0 & 2 & 6 \\ 3 & 6 & 2 & 0 & 2 \\ 4 & 2 & 6 & 2 & 0 \\ & 1 & 2 & 3 & 4 \end{bmatrix}$$

$$D_{cos\theta} = \begin{bmatrix} 1 & 0.71 & 0.76 & 0.98 \\ 0.71 & 1 & 0.98 & 0.76 \\ 0.76 & 0.98 & 1 & 0.83 \\ 0.98 & 0.76 & 0.83 & \times \end{bmatrix}$$

0.98, 0.83, 0.76
① ③ ②
Y N N
k = 1 → Y
k = 2 → Y
k = 3 → N

$\vec{1} = [1, 2, 3]$
$\vec{2} = [3, 2, 1]$

$$cos\theta = \frac{1 \cdot 2}{\|\vec{1}\|\|\vec{2}\|} = \frac{3+4+3}{\sqrt{14}\cdot\sqrt{14}} = \frac{10}{14} = \frac{5}{7} \approx 0.71$$

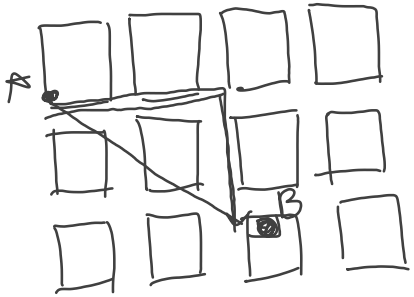
$$\frac{\vec{1}\cdot\vec{1}}{\|\vec{1}\|\|\vec{1}\|} = \frac{1+4+9}{\sqrt{14}\ \sqrt{14}} = \frac{14}{14} = 1$$

Euclidean A



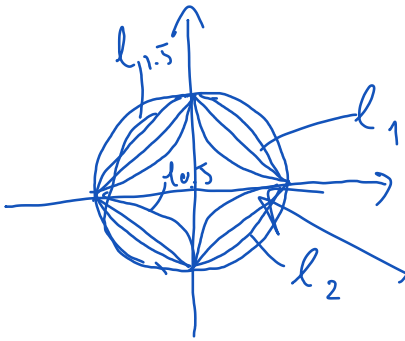
$$\ell_2\text{-norm} = \sqrt{(a_1-b_1)^2+(a_2-b_2)^2} = \left(|a_1-b_1|^2+|a_2-b_2|^2\right)^{\frac{1}{2}}$$

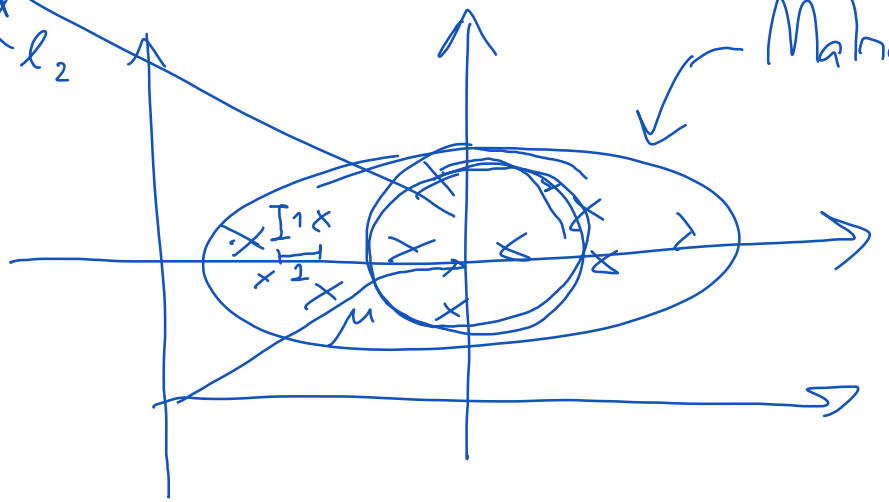
Manhattan

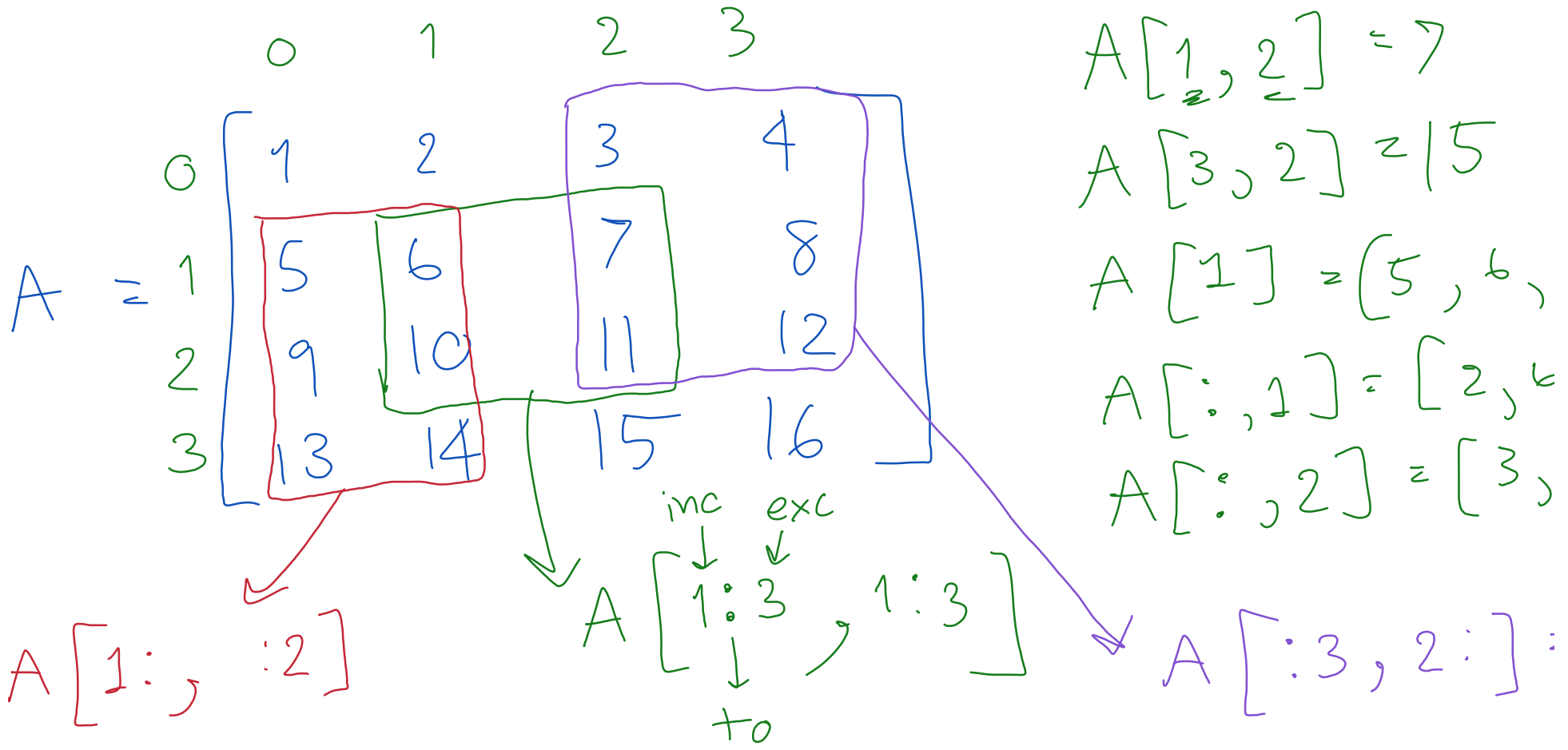


$$\ell_1\text{-norm} = |a_1-b_1|+|a_2-b_2| = \left(|a_1-b_1|^1+|a_2-b_2|^1\right)^{\frac{1}{1}}$$

$$\ell_p\text{-norm} = \left(|a_1-b_1|^p+|a_2-b_2|^p\right)^{\frac{1}{p}}$$






A =
0 1 2 3
0 1 2 3 4
1 5 6 7 8
2 9 10 11 12
3 13 14 15 16
A[1:, :2]
A[1:3, 1:3]
inc
exc
to
A[:3, 2:]
A[1, 2] = 7
A[3, 2] = 15
A[1] = (5, 6,
A[:, 1] = [2, 6
A[:, 2] = [3,

:2 = 0, 1

:3 = 0, 1, 2

$2:4 = 2,3$

$2:8:2 = 2,4,6$

start end step

$-1 = end$

$-2 = end-1$

$-2: = end-1, end$

$\begin{bmatrix} 1 & 2 & 5 \\ 3 & 4 & 6 \end{bmatrix}$ -